Professional Ethics and Human Values

Ethics is a necessary and listed Graduate Attribute for all engineers according to the Washington Accord. As engineers deal with society and provide for society, it is important that the ethical concerns pertaining to technology are well understood and addressed, particularly when technological advancements are still in their infancy and early growth stages but already enjoy a widespread use! Human Values form the basis for all ethics, and ethical theories help resolve professional dilemmas too. This course aims to create an appreciation for normative and applied ethics with special focus on professionalism as well as technology education and practice. Given the diverse set of roles a STEM professional may play in a society, there is an inherent societal need for STEM professionals to be ethical. The formative years of students of engineering are deemed the best time to impress upon them the practical importance and application aspects of ethics. The curriculum is designed to include an inherent appreciation for the Indian Ethos and cover a wide variety of topics with suitable case studies and examples all through. The course is aimed to become a practicable guide that enables the learners to apply the learned content while facing the practical contexts encountered in their individual life and career, even as the times and lives are ever-changing.

As a course designed to ideally educate and empower all engineers, this course content contributes substantially to the attainment of "Ethics" as one of the eleven Graduate Attributes identified in the Washington Accord, and contributes moderately to a few other Graduate Attributes—including but not limited to The Engineer and The World, Lifelong Learning, and Design/Development of Solutions—while also contributing slightly to the attainment of some more Graduate Attributes such as Individual and Collaborative Team Work, Communication, Project Management and Finance.

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. V Semester

(22OE1HS301) PROFESSIONAL ETHICS AND HUMAN VALUES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To emphasise on the importance of ethics for STEM professionals
- To help create a personal and practicable toolkit for ethical behaviour in personal and professional settings
- To relate the profession of engineering to sociocultural, ethical, and moral contexts in India and globally
- To develop more socially conscious STEM professionals who create and conceive a better society and a better world without sacrificing or ignoring public good

COURSE OUTCOMES: After completion of the course the students should be able to

CO-1: Distinguish morals, values, and ethics in Indian and global contexts as well as personal, sociocultural, and professional contexts

CO-2: Resolve moral and ethical dilemmas through relevant ethical guidance using ethical inquiries and/or appropriate ethical theories

CO-3: Demonstrate understanding of professionalism and be responsible social experimenters as STEM professionals in diverse roles

CO-4: Be conscientious about professional contributions that adhere to digital and technology ethics principles in ensuring public welfare and sustainable development

CO-5: Extend and apply individual moral/ ethical persona in organisational, business, and global environments

UNIT-I:

Introduction and Motivation

Definitions: Morals, Values, Ethics; Golden Rule; Maslow's Theory of Needs; Diversity, Equity, Inclusivity, and Accessibility (DEIA); Sustainable Development Goals (SDG)

Human Values: Universal Human Values, Conventional and Constitutional Values in Indian Ethos, Building Blocks for Moral Persona—Conscientiousness, Integrity, Empathy Morality and Moral Development: Moral Character, Moral Autonomy; Platform vs. Purpose

Ethicality: Civic Virtue for Ideal Society vs. Anomie; Ethics as basis for Legal Framework; Indian Ethos

Motivation: Examples and Case Studies of Ethical and Moral Decline in social, business, and professional contexts; Threats to Ethics due to changing technology landscape; Need for protection of Personally Identifiable Information (PII) and Data Privacy; Introduction to Trolley Problem and Discussion on Moral and Ethical Dilemmas; Issues in Decision-making—Value Judgement and Moral Judgement; Ethics in Academics and Research

UNIT-II:

Human Values, Moral Development, and Ethical Theories of Right Action

Moral Development: Kohlberg's Theory vs. Gilligan's Theory; Understanding Biases

Ethical Theories: Classification of Ethical Theories—Prescriptive, Descriptive, and Applied Ethics by Character and by Conduct; Deontological Ethics (Immanuel Kant, John Rawls), Virtue Ethics (Aristotelianism), Consequentialism (Act & Rule Utilitarianism); Divine Command Ethics and Ethics through Spirituality; Ethical Pluralism; Overview of Rights Theories

Ethical Guidance: Ethical Frameworks and Values for organisations; Practicable guidance from the Bhagavad-Gita and Eastern Philosophies; Ethical Inquiries—Normative, Conceptual, and Factual; Ethical Decision-making Tools; Ethics and Law—Ethics as a Basis for Law, Awareness of Rights of Convicts and Criminals

UNIT-III:

Ethics for STEM Professionals

Profession and Professionalism: Definitions; Professional Characters, Traits, Virtues, Roles, Rights and Responsibilities; Overview of Business Ethics and Work Ethics; STEM Professionals as Managers, Leaders, Consultants, Expert Witnesses, Scientists, Researchers and Academicians, and Entrepreneurs

STEM Profession as Social Experimentation: Comparison with scientific experiments; Application of Ethical Theories and Ethical Inquiries in professional contexts to gather relevant information; Informed Consent

Responsibilities of STEM Professionals as Social Experimenters: Need for being Ethical by Design; Accountability and Answerability in general and in autonomous technology applications; Focus on DEIA, SDG, and Social Innovation; Guidance through Professional Societies

UNIT-IV:

Understanding Ethics through STEM Case Studies

Safety and Risk: Understanding Safety and Risk through case studies—Titanic, Chernobyl, Bhopal; Overview of risks implied by digital technologies

General Aspects: Environmental Ethics and Carbon Footprint; Non-disclosure Agreements (NDA) and Intellectual Property (IP) as a Professional Right and as exploited by businesses; Copyrights and Ethics in Arts; Social Engineering and Ethics

Digital Ethics: Data Ethics; Ethics of AI; Ethics on Social Media; Ethics of Humans as Data Objects; Legal frameworks and policies for Data Management—Content Development, Protection, Integrity, Transparency, and Accountability; Ethics in the context of emerging technology areas such as Large Language Models (LLM), Digital Forensics, Internet of Things (IoT), Robotics and Automation, Cybersecurity & Cybercrime, Healthcare & Clinical Management, Finance, and Personal Tech

Technology Ethics: Computer Ethics, Machine and Robot Ethics; Ethics of technology advancements — an ethical urge to create green technologies, steps taken in the EU

UNIT-V:

Ethics for Workplace and Societal Wellbeing

Ethics in Organizational Behavioral Aspects: Collegiality, Loyalty, Trust; Workplace Deviances, Occupational Crime, and Culpability; Employee Rights and Relations; Ethical Frameworks and Codes of Ethics and Conduct; Conflicts of Interest, Conflict

Management and Resolution; Creating Ethical Climate, Ethics Education and Training, Grievance Redressal Procedures; Whistleblowing

Global Context: Work-Life Balance, DEIA at workplace, Cultural aspects, Concerns for Multinational Companies

Ethics and Society: Impact of Wars, Politics, and Religion on society; Understanding Privilege, Marginalisation, and Minorities; Ethico-economical Aspects: Fair Trade, Capitalism vs. Communism, Developed vs. Developing vs. Underdeveloped economies

TEXTBOOKS:

1. Ethics in Engineering, Mike W. Martin, Roland Schinzinger, McGraw Hill Education, 2017 (ISBN: 978-9339204457)
2. Business Ethics: An Indian Perspective, A.C. Fernando, K.P. Muralidheeran, E.K. Satheesh, Pearson Education, 2019 (ISBN: 978-9353437442)
3. Professional Ethics, R. Subramanian, Oxford University Press, 2017 (ISBN: 978-0199475070)
4. Professional Ethics and Human Values, M. Govindarajan, S. Natarajan, V.S. Senthil Kumar, Prentice Hall India, 2015 (ISBN: 978-8120348165)

REFERENCES:

1. Engineering Ethics: Concepts & Cases, Charles E. Harris, Jr., Michael S. Pritchard, Michael J. Rabins, Cengage Learning, 2012 (ISBN: 978-8131517291)
2. Classical Indian Ethical Thought: A Philosophical Study of Hindu, Jaina and Bauddha Morals, Kedar Nath Tiwari, Motilal Banarsidass Publishers, 2017 (ISBN: 978-8120816084)
3. Ethics for the Whole World, Dalai Lama, 978-9351360803 Vijay Kumar Ivaturi et al., The Manual for Indian Start-Ups, Penguin Random House India, 2017 (ISBN: 978-0143428527), To Be Human, Jiddu Krishnamurti, Shambhala, 2000 (ISBN: 978-1570625961)
4. On Ethics and Economics, Amartya Sen, Oxford India, 1999 (ISBN: 978-0195627619)